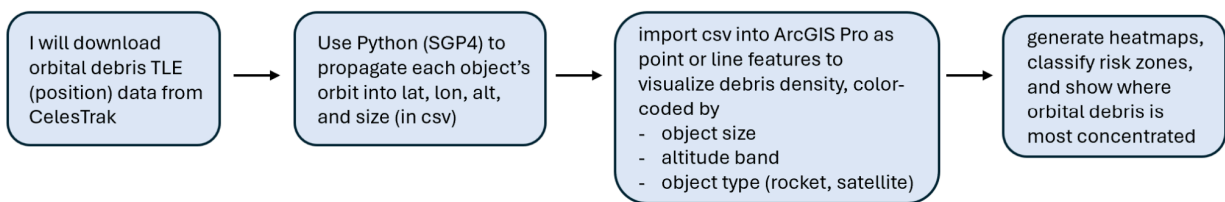


Project description: Global Spatial Distribution of Orbital Debris: A GIS-Based Density Analysis

Problem definition: Any nonfunctional human-made object that is in earth's orbit or reentering the atmosphere, from four-ton rocket bodies to millimeter-wide pieces of metal, can be classified as orbital debris. As launch activity accelerates and the number of objects placed in orbit continues to grow, orbital debris is emerging as an increasingly serious threat to the sustainable use of space, which is crucial to the United States for communication, national security, transportation, and environmental monitoring purposes.

Approach:



Datasets: 2026 Population data: [WorldPop :: Population Counts](#)

Catalog of debris positions: [Celestrak: Search Satellite Catalog](#)

Orbits (csv): exported after python data analysis, input is Celestrak data

Global launch sites: [Rocket Launch Sites by Kaggle](#) | [Other](#) | [Baselight](#)

Desired Results: I hope to visualize the extremely high density of orbital debris through mapping the groundtracks (projecting the orbits of the debris onto the earth) and also classifying it by size (for example, coloring the groundtracks by the size of the debris). I will also map population density to show how the debris could potentially impact highly populated areas or launch sites.

References:

[Cost and Benefit Analysis of Mitigating, Tracking, and Remediating Orbital Debris](#)

[NASA SVS | Tracking Satellites and Space Debris in Earth Orbit \(Feb 2024\)](#)